

Memorandum

To: EAGLE Engineering / NCI Office of Acquisitions **From:** EAGLE Platform Team **Date:** 2026-05-05 **Subject:** Document Type Registry — Single Source of Truth Refactor

Background

EAGLE generates 24 distinct federal acquisition documents through `create_document` (SOW, IGCE, J&A, QASP, Section 889 Compliance, etc.). The canonical document-type list had drifted across multiple backend layers — labels, prompts, generators, templates, aliases, and the compliance map each held their own copy. Adding a new document type required keeping all of them in sync, and any miss produced runtime "Unknown document type" errors or silent registry conflicts.

The triggering incident: a contracting officer's package required QASP, HHS-653 Small Business Review, Required & Priority Sources Checklist, and Section 889 Compliance documents. The package compliance map demanded those slugs but the document generator rejected them — the two layers had drifted out of sync over multiple commits. Manual investigation of the diagnostics output identified four additional class-A blockers (D&F, Source Selection Plan, Human Subjects Provisions, COR Designation Letter) and four others requiring different routing (SAM exclusions, FPDS report, Inherently Governmental Certification, Wage Determination).

Action Taken

A single-source-of-truth refactor was implemented to make this class of drift structurally impossible. All slug-keyed behavior — labels, LLM system prompts, generators, S3 templates, alias resolution, and compliance-matrix integration — now derives from one declaration per slug in `server/app/doc_registry.py`. A module-load validator runs at import time and refuses to start the server if any inconsistency is detected.

The refactor consolidated:

- 23 per-slug generator wrapper functions (deleted)
- The 600-line `_DOC_TYPE_SYSTEM_PROMPTS` dict (relocated to `doc_prompts.py`, derived from registry at runtime)
- The 200-line `TEMPLATE_REGISTRY` dict literal (replaced with a builder that reads the registry)
- The duplicated `DOC_TYPE_ALIASES`, `_FALLBACK_DOC_TYPES`, `_MARKDOWN_ONLY_TYPES`, `_DOC_TYPE_LABELS`, `FORM_TEMPLATES`, and `_COMPLIANCE_DOC_TO_SLUG` dicts (all now derived views)

In addition, **8 previously-blocked document types** were added to the registry and wired end-to-end: QASP, HHS-653 SB Review, Priority Sources Checklist, Section 889, D&F, Source Selection Plan, Human Subjects Provisions, COR Designation Letter, Inherently Governmental Certification, and Wage Determination. Two additional types (SAM exclusions, FPDS report) were classified as `EVIDENCE` kind — externally fetched data routed away from `create_document` with an explicit error pointing operators to the package evidence path.

Results

Metric	Before	After
Adding a new doc type requires	Edits across multiple modules in sync	One registry entry
Slugs in the registry	27	44
Slugs accepted by <code>create_document</code>	23	41
Drift detection	Runtime errors only	Module-load validator
Tests passing	162/162	162/162
Lint clean	yes	yes

The validator catches alias collisions, slug-key mismatches, generated documents missing both prompt and template, and evidence-kind documents incorrectly declaring a prompt or template. None of these conditions can now reach production.

Remaining Template Gaps

17 generated document types still produce raw markdown rather than filling an official S3 template. Eight of them have known federal/HHS forms and would benefit from template wiring:

Slug	Template source	Priority
<code>cor_designation</code>	Reuse existing NIH COR Appointment Memorandum	Quick win — template already in S3
<code>sb_review</code>	HHS-653 form (slug is named after it)	High
<code>section_889</code>	SAM.gov 52.204-26 attestation	High
<code>section_508</code>	GSA VPAT/ACR template	High
<code>priority_sources_checklist</code>	HHS/NIH "GSA First" PDF	High
<code>subk_review</code>	HHSAR 352.219 fillable checklist	Medium

Slug	Template source	Priority
qasp	NIH PBA handbook example	Medium
source_selection_plan	NCI/HHSAR 315.3 SSP template	Medium

Six additional types (D&F, Contract Type Justification, Price Reasonableness, Required Sources, Inherently Governmental, Eval Criteria) could be templated with lower expected gain. Three (Wage Determination, Human Subjects, Security Checklist) are best left as markdown — they are either data lookups or system-specific narratives that resist a single template.

Recommendation

Begin with `cor_designation`. The NIH COR Appointment Memorandum docx is already in S3 and currently used by the related `cor_certification` slug; pointing `cor_designation` at it is a roughly 15-minute change that exercises the registry's "add a template = edit one file" property. From there, prioritize `sb_review` and `section_889` since both are explicitly named after federal forms that compliance reviewers expect to receive in the standard layout.

The remaining markdown-only types should not be templated speculatively. Each new template carries an ongoing maintenance cost (placeholder map, fallback handling, test coverage) and should be justified by either a compliance auditor request or a documented operator complaint about output format.

Files of record:

- Spec: `.claude/specs/20260505-155739-plan-doc-registry-ssot-v1.md`
- Registry: `server/app/doc_registry.py`
- Prompts: `server/app/doc_prompts.py`
- Migrated consumers: `ai_document_schema.py`, `doc_type_registry.py`, `template_registry.py`, `package_store.py`, `tools/document_generation.py`, `tools/create_document_support.py`

Validation: `ruff check` clean across all 8 modified files. 162/162 unit tests pass (`tests/test_template_schema.py`, `test_unify_document_tracking.py`, `test_dynamic_required_docs.py`, `test_ai_document_schema.py`, `test_template_service.py`).

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